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A PENNSYLVANIAN PALEOMAGNETIC POLE FROM THE MINERALIZED LATE CAMBRIAN BONNETERRE FORMATION, SOUTHEAST MISSOURI

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Abstract. One hundred eighty-eight oriented samples of the Late Cambrian Bonneterre carbonates of southeastern Missouri were taken from Viburnum Trend mines and local surface outcrops. Initial natural remanent magnetization directions have south-southeasterly declinations and downward inclinations. Progressive demagnetizations using thermal and AF techniques give two directions: one is a viscous present-day field direction, and the other is of reversed polarity, with south-southeasterly declinations and very shallow inclinations. The resulting paleopole, at 43°N, 126°E, is located near the Pennsylvanian to Early Permian segment of the North American apparent polar wander path. A negative conglomerate test supports this assignment for the magnetization age. Magnetic extracts from the mineralized as well as unmineralized zones of the Bonneterre contain spherical crystal aggregates of pure magnetite of late diagenetic origin. The secondary magnetization resides in the magnetite spheroids, and we infer that the spheroids formed prior to sulfide mineralization.

Introduction

Diagenetic processes occurring in intracratonic basin environments are difficult to date by either stratigraphic or geochronological methods. The Cambrian Bonneterre Formation of southeastern Missouri presents an example of a carbonate sequence that underwent multiple stages of diagenesis, including mineralization of epithermal Mississippi Valley Type (MVT) deposits. Study of these deposits since the early 1800's has not solved the fundamental problem of the age of the MVT deposits, and the origin of the mineralizing fluids remains obscure. Relative age relationships of ore and gangue mineral deposition can be derived through careful study on a local scale, but determining limits on the absolute age of regional mineralization has thus far proved futile. Structural relationships between collapse breccias and interstitial lead ores allow for possible ages encompassing most of the Paleozoic, and U/Pb isotope methods give anomalous dates. This study examines the paleomagnetism and magnetic mineralogy of the Bonneterre Formation and some of its ore deposits in an attempt to bracket the age of the mineralization.

Only reconnaissance paleomagnetic studies of the Bonneterre Formation have been carried out to date, chiefly in the subsurface mines of the Viburnum Trend, and the overall paleomagnetic behavior of the Bonneterre is not well known. Beales et al. [1974] investigated MVT deposits in the Cambrian strata of Newfoundland and also southeastern Missouri and interpreted the paleomagnetic directions obtained through alternating field (AF) cleaning as representing Late Cambrian North American poles. They suggested that the original magnetization (and mineralization) had occurred soon after deposition of the host rock. However, more recently, Wu and Beales [1981] focused on the Viburnum Trend and a few nearby surface sites and found poles falling on the better known Pennsylvanian-Permian segment of the North American apparent polar wander (APW) path, although directions from a few surface samples geographically removed from this trend were inferred to correlate better with Cambrian poles. Remagnetization of the Bonneterre by magnetic mineral rotations caused by matrix dissolution and recrystallization was inferred by Wu and Beales from Verosub's [1977] review of such processes. Neither study employed extensive demagnetization techniques or statistically significant numbers of samples.

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Geologic Setting

The geology of southeastern Missouri has been described by Snyder and Gerdemann [1968] and Thacker and Anderson [1977] and is briefly summarized here. The Bonneterre carbonates are part of a cratonic Cambro-Ordovician sequence of marine transgressive rocks exposed around the Ozark Dome in southeastern Missouri (Figure 1). The uplifted rocks are mostly Precambrian rhyolites and granites. Continent-wide inundation in the Late Cambrian is represented first by the Lamotte sandstone. Isolation from clastic sources, upon further submergence, allowed the development of the Dresbachian Bonneterre carbonates around the exposed Precambrian core. Dolomitization of the Bonneterre is thought to have occurred soon after lithification by introduction of fluids from an outside source [Lyle, 1977]. The overlying Davis Formation consists of shale units interbedded with dolomites, possible evidence of further sea-level fluctuations. Through the rest of the Late Cambrian and up to the Middle Ordovician, massive carbonates interbedded with thin sandstone units were laid down until deposition was terminated by major regressions, creating the Saukian unconformity.

Carbonate facies in the Bonneterre are distributed symmetrically about the Ozark Dome and consist of low-energy planar stromatolites and burrowed lime muds near the dome surrounded by digitate stromatolite reef complexes and subtidal oolitic sands. Beyond lies a forereef facies of micritic dolomites and shales. There is a limestone/dolomite transition below the subtidal facies (Figure 2). The reef facies, which contains the main bodies of lead ore, reached its maximum horizontal extent during a regression partway through its section. A Late Cambrian transgressive sequence onlaps the back reef and Precambrian rocks, resulting in the disappearance of the digitate stromatolite reef as the oolite facies expanded into both reef and forereef environments. Unique porosity or permeability properties of both the digitate stromatolite reef and the oolite facies probably enhanced later karstification, which selectively prepared those facies for ore deposition.

As in most of the MVT mining districts or North America, the age of mineralization in the Viburnum Trend is not firmly constrained by evidence of any kind

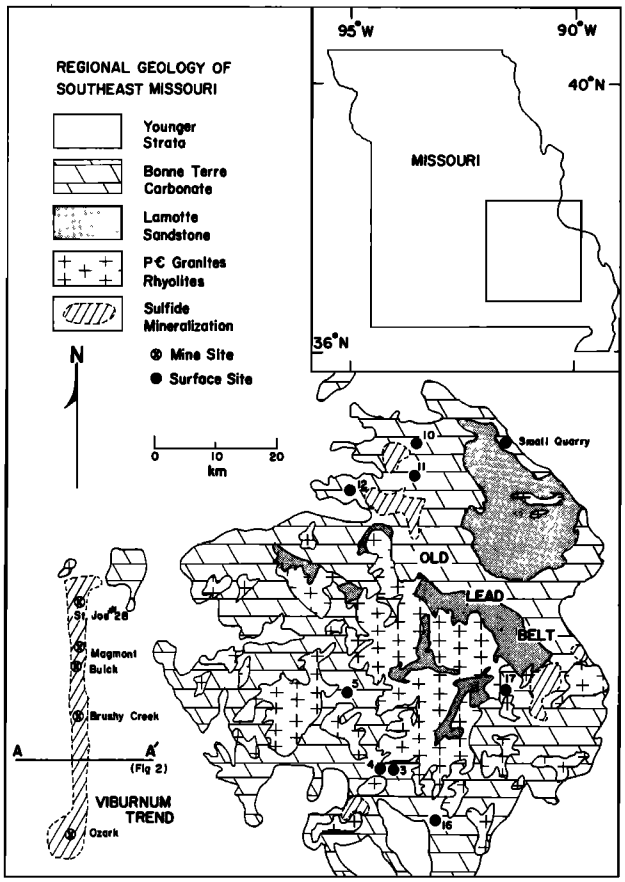


Fig. 1. Regional geology of the Late Cambrian strata of the Ozark Dome, southeast Missouri [Anderson, 1979], with sampling localities indicated. Note position of cross-section AA' represented in Figure 2.

other than that the epigenetic ores must be younger than lithification of their host rocks. A significant fraction of the mineralization is located in breccias of probable karst-related, solution collapse origin, but the age of such brecciation is debatable: an established period of regional emergence and widespread erosion in Middle Ordovician (post Knox) time [Thacker and Anderson, 1977] seems to be a reasonable, but nevertheless unproven, time for development of this particular type of ore-controlling breccia. In a thoughtful analysis of this problem as it applies in the southeast Missouri lead district, Snyder [1967, p 12] was forced to conclude that 'there is no evidence to indicate time of mineralization, although it did not occur prior to post-lithification faulting, probably mid-Ordovician or later.' Attempts at radiometric dating have been equally inconclusive. As summarized by Heyl et al. [1966], the highly radiogenic (J-type) leads of the district give anomalous ages for galena deposition. More recently, York et al. [1980] have attempted direct dating of the ore minerals of southeastern Missouri by using the $^{39}Ar/^{40}Ar$ variant of the K-Ar method, but their preliminary results seem questionable. They obtain ages of 700 and 380 Ma for pyrite and galena, respectively, in the same ore specimen. While it seems most unlikely that two sulfides of such diverse age would be similarly located in a single ore deposit, more serious questions are attached to the significance of argon 'isochrons' derived by this method. The source of potassium and argon in such measurements must reside in fluid inclusions in these sulfides, and the apparently time-dependent isochrons may merely be expressing mixing trends between end member fluids of different K and Ar contents.

In summary, past efforts at geological and isotopic dating of the MVT ores of Missouri have produced, at best, subjective or inconclusive results. Ohle [1980] was the latest to stress that, until MVT ore deposits can be accurately dated, this important type of mineralization is difficult to interpret in genetic terms. Thoughts of this kind motivated the present study in the Viburnum district.

Sampling and Laboratory Treatments

A total of 188 samples of the Bonneterre was taken at 25 sites at the surface around the Ozark Dome and in the mines of the Viburnum Trend. The distribution of sites and facies can be seen in Figure 1. The subsurface samples are important because of the effects of weathering on surface samples, which will be described later. Sampling of large breccia blocks (0.5 -3.0 m) and associated slump folds in the mines provided conglomerate and fold tests unavailable at the surface.

Using a portable rock drill, 117 oriented samples were taken from nine nearly flat-lying surface sites. Road cuts and river-cut cliffs offered up to 15 m of exposure at each site. The surface sites are distributed around the Precambrian core of the Ozark Dome (Figure 1), but tend to lie within the main reef facies and ore trends. Only the Small Quarry site to the east is removed from the reef complex.

We gained permission from five of the major mines along the Viburnum Trend to take 71 oriented hand samples from the subsurface mineralized zone (Figure 1). Ten to eighteen samples were taken at each of several widely distributed sites within each mine. Many samples from the heavily mineralized zones contained galena and marcasite-pyrite gangue, and only a few rock samples appeared unaffected by mineralization. Breccia blocks were sampled from the ore trends in the Brushy Creek and St. Joe #28 mines for conglomerate tests. Gentle slump folds caused by local solution collapse were also sampled in the Buick Mine for a fold test. In addition to the subsurface samples taken from the mines, a number of unoriented drill core samples

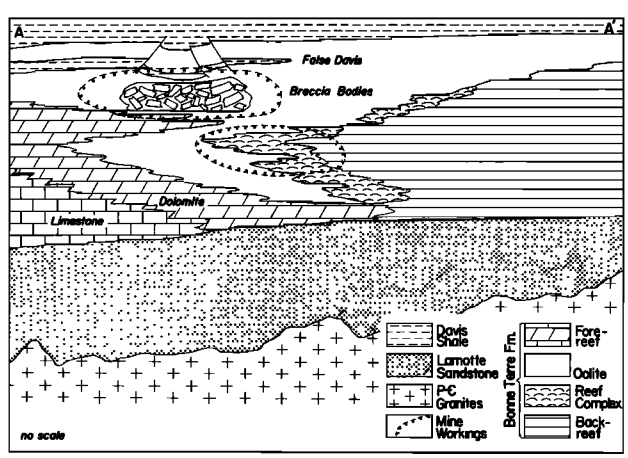


Fig. 2. Stratigraphic cross-section of the Viburnum Trend (see Figure 1). A general reef facies progression is evident from east to west, along with the dolomitization front beyond the reef [after Larsen et al., 1979].

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TABLE 1. Paleomagnetic Site Means and Formation Mean

Site	Location, deg	Decl/Incl, deg	N/N_o	k	α_{95}	R	Pole position, deg
Viburnum Trend							
St. Joe #28 Mine	37.72 N, 91.12 W	154/+ 2	18/32	12	10.2	16.62	44 N, 128 E
Magmont Mine	37.65 N, 91.12 W	157/+ 4	16/24	31	6.8	15.51	45 N, 123 E
Buick Mine	37.62 N, 91.12 W	149/+ 5	25/26	22	6.3	23.91	40 N, 132 E
Brushy Creek Mine	37.55 N, 91.13 W	153/+11	28/32	16	7.0	26.34	40 N, 128 E
Ozark Mine	37.37 N, 91.14 W	155/+ 1	38/38	23	4.9	36.40	46 N, 126 E
Surface Sites							
BT - 11	37.90 N, 90.48 W	153/+14	9/15	13	14.8	8.39	38 N, 124 E
BT - 16	37.34 N, 90.41 W	149/+ 4	15/17	60	5.0	14.76	41 N, 133 E
BT - 17	37.58 N, 90.35 W	164/+15	3/ 4	11	38.5	2.82	42 N, 111 E
Formation Mean		154/+ 7	8/ 8	120	5.1		43 N, 126 E

N/N_o is the ratio of entries used in the statistical analysis to the number of samples studied; k and α_{95} are the statistical parameters [Fisher, 1953] associated with the mean and R is the resultant vector length.

taken from 50 to 100 km from the Ozark Dome were supplied by the Missouri Geological Survey.

All samples were cut into 2.5 cm diameter by 2.1 cm long cores, and their natural remanent magnetization (NRM) was measured with a ScT cryogenic magnetometer. This instrument has a sensitivity of better than 1×10^{-5} A/m, which is adequate for the study of carbonates. Stepwise thermal and AF demagnetizations were carried out to isolate the components of the NRM. Orthogonal demagnetization diagrams and vector subtraction [Zijderveld, 1967] were used to calculate directions of magnetization, and paleomagnetic poles were calculated for comparison with the North American APW path. Rock magnetic experiments and analysis of magnetic extracts allowed us to characterize the magnetic mineralogy of the samples and to infer the origin of the magnetizations.

Results

Paleomagnetism

Of the nine surface sites sampled, only three showed any consistent behavior during demagnetization. Many of the samples from the sites that did not yield consistent results exhibited random changes in magnetic directions and intensities during demagnetization. A light brown color of the rocks from these sites was occasionally observed in the field and suggested recent weathering. Isothermal remanent magnetization (IRM) acquisition and demagnetization behavior, described below, suggested the presence of two magnetic phases. Sites 3 and 4 yielded poor results because of extremely low NRM intensities.

Samples from three surface sites (11, 16, 17) behaved consistently during thermal and AF demagnetization (Table 1). Initial NRM intensities averaged 2.5×10^{-4} A/m; we considered measurements unreliable when intensities fell below 1×10^{-5} A/m during demagnetization. Typical demagnetization diagrams (Figures 3a, b) show removal of a magnetization aligned with the present-day geomagnetic field of probable viscous origin below about 20 mT or 300°C. Further demagnetization reveals a characteristic magnetization with south-southeasterly declinations and very shallow inclinations that was almost completely removed by 100 mT or 580°C.

Several sites, up to a few kilometers apart, were sampled in each of the Viburnum Trend mines. Dolomitization and mineralization is of the same general nature and intensity at similar stratigraphic levels in all the mines, and no spatially or stratigraphically controlled difference in magnetic behavior was found. Oxidation effects or major post-mineralization diagenetic processes were not noted, and all samples appeared unweathered.

Directions obtained from thermal and AF demagnetizations of these mine samples were generally much better defined and showed more consistency than those from the surface sites (Figure 3c, d). However, behavior during demagnetization is very similar and little difference can be seen between surface (Figure 3a, b) and subsurface (Figure 3c, d) directions. Most of the directions are of reversed polarity (D/I = 154°/7°), but eight normal polarity directions, approximately antiparallel to the reversed ones (D/I = 323°/1°), were found in the Ozark Mine samples. The characteristic directions from both mine and surface samples cluster well (Figure 4) with k = 120 and alpha-95 = 5.1° [Fisher, 1953]. The resulting paleopole lies at 43°N, 126°E (Figure 5).

Stability tests were carried out by using structural relationships. When the main breccia bodies formed, adjacent material tended to detach along listric normal faults and slide into the breccia bodies [Larsen et al., 1979]. Gentle folds that developed within the moving blocks have approximately north-south fold axes and fold limbs dipping up to 15° and extend for 100 m or so. A fold test [Graham, 1949] was attempted from a well-exposed fold system in the Buick Mine but was inconclusive because of the small angle between the fold axes and the magnetic direction (Table 2).

Well-preserved, internally bedded breccia blocks offered an opportunity to test the magnetic stability of these carbonates with a conglomerate test [Graham, 1949]. Characteristic directions from breccia samples

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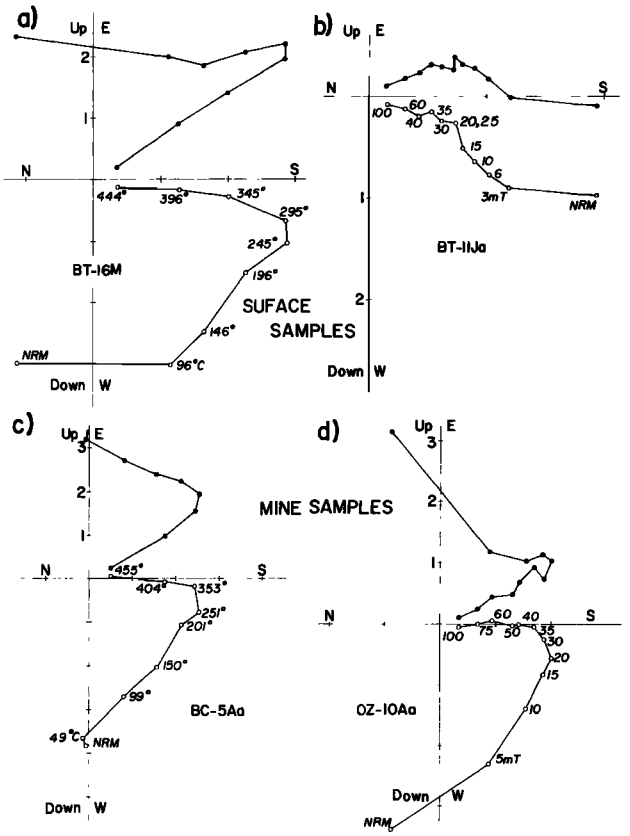


Fig. 3. Typical demagnetization diagrams for the Bonneterre Formation. Closed circles represent projections onto the horizontal plane at various stages of demagnetization, open circles represent projections onto the vertical plane. Intensities are in units of 10^{-4}A/m. (a) Thermal demagnetization of surface sample from site 16. (b) Alternating field demagnetization of a surface sample from site 11. (c) Thermal demagnetization of mine sample from the Brushy Creek Mine. (d) AF demagnetization of mine sample from the Ozark Mine.

from the St. Joe #28 Mine (Figure 6, Table 2) cluster significantly better before a tilt correction is applied [McElhinny, 1964], which shows that the magnetization was acquired after brecciation. In contrast, results of conglomerate tests from the Brushy Creek Mine (Table 2) are similar to those of the fold test from the Buick Mine in that the change in directions upon tilt correction is not significant. Although a small decrease in the precision parameter (k) was observed, the conglomerate test is inconclusive in this case.

Two other sets of samples were gathered to test hypotheses on the extent and regional characteristics of this secondary magnetization. Wu and Beales [1981] previously sampled the Small Quarry in the Bonneterre east of the mineralized zones. They found, after thermal cleaning, an east-southeasterly and shallow direction that they interpreted to be Cambrian. From the eight samples we collected there, no such direction was isolated by thermal demagnetization. Only the present-day field direction was observed. Several unoriented but vertical deep sample cores were acquired from the Missouri Geological Survey from wells 50 to 200 km to the NW and SW of the Ozark Dome. Samples were demagnetized in search of shallow inclinations. Only unusually strong and vertically down magnetizations could be found, possibly induced by the action of prolonged drilling.

Rock Magnetism

From the demagnetization studies, some properties of the magnetic carriers can be inferred. The present-day field component is characterized by blocking temperatures up to 300°C and coercivities of about 25 mT. It is likely that the minerals carrying this component have a short relaxation time, possibly being low-coercivity magnetite [McElhinny, 1973, p. 55]. The characteristic component is eliminated below 600°C and is largely removed below 100 mT. This indicates magnetite or possibly low-coercivity hematite.

Acquisition of IRM was studied in samples chosen from a variety of lithologies and stratigraphic positions. Two different types of behavior were found (Figure 7). All samples taken from the subsurface (Viburnum Trend samples, unoriented deep core samples) quickly acquired an IRM below 0.3 T, upon which saturation was reached and intensities remained constant up to 5.0 T. Such behavior is consistent with that of magnetite [Parry, 1965]. In contrast, the surface samples, after showing similar behavior up to 0.3 T , continued to acquire an IRM with increasing fields, but at a much slower rate. Saturation was not reached, even in fields as high as 5.0 T, indicating that hematite and/or goethite are present [Dunlop, 1971].

Low-temperature studies [Mauritsch and Turner, 1975] were also carried out. A representative sample was cooled to liquid nitrogen temperatures (-196°C) and given an IRM, and the magnetization was measured in the cryogenic magnetometer as the sample warmed, using a method described by McCabe et al. [1982]. Magnetization was plotted as a function of temperature to see if the low-temperature magnetocrystalline transition was present. A steady decrease in magnetization was noted as the sample warmed, but no well defined transition is present. This result does not preclude magnetite but suggests that if magnetite is present, anisotropies other than the magnetocrystalline one may predominate.

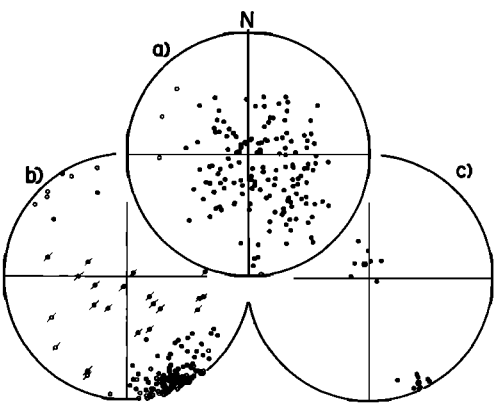


Fig. 4. Equal-area projections of Bonneterre magnetizations. Solid circles indicate positive inclinations, open circles represent negative inclinations. (a) Natural remanent magnetizations of mine samples. (b) Characteristic directions obtained from demagnetization of all samples. Slashed symbols represent directions excluded from the means. (c) Site-mean, present-day field, and characteristic directions of the Bonneterre. Hexagons represent the overall means of these directions.

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Magnetic Extracts

From IRM intensities we estimate an average magnetite concentration of about 2 ppm in the samples. Given these low concentrations, thin section searches to locate and characterize iron minerals would have been time consuming and probably futile. To circumvent this problem, magnetic extracts were obtained for direct observation of the magnetic fraction. About 0.1 kg of representative rock samples were powdered by hand to approximately 3 phi units size and suspended in distilled water. This suspension was then either pumped slowly past an electromagnet or exposed to the field of a small hand magnet outside a holding beaker. A small amount of extracted magnetic material was removed and prepared for scanning electron microscope (SEM) and X ray diffraction studies.

An extract from a mine sample contained about 50 magnetic spheroids and an equal amount of magnetically responsive dolomite and quartz crystals (Figure 8). The spheroids range in size from a maximum of 156 micrometers to about 5 micrometers in diameter. The larger spheroids have relatively smooth surfaces and appear to be aggregates of crystals several micrometers across (Figure 8a, b). Several broken spheroids with hollow cores were observed (Figure 8c). The inner surfaces are covered with protruding anhedral crystals similar in size to those on the outer surface (Figure 8d). Qualitative energy-dispersive analyses of the spheroids show iron as the only cation present in detectable amounts, and no compositional zoning was observed in the broken spheroids.

Studies of extracts from subsurface limestone cores taken far away from the mineralized zones also revealed a few small iron oxide spheroids (though not nearly as many or as varied as those from the mines) and a few 5 micrometer diameter pyrite framboids. Unfortunately, dolomite was also extracted, and this obscured the search for possible magnetic grains. Extracts from surface sites 11 and 16 revealed several spheroids of up to 40 micrometers in diameter that exhibited surface textures similar to the spheroids from the mines. As with the iron oxide spheroids from mine samples, those from the cores contained only iron (no Ti, Ni, or S) in detectable quantities.

A powder X ray diffraction pattern was obtained from the mine sample extract by using a Gandolfi camera and Fe K-alpha radiation. The pattern showed that the extract contained a spinel, and the least squares refined lattice parameter is consistent with pure magnetite. These results agree with the energy dispersive analytical data that indicated a pure iron mineral, probably an oxide (no sulfur present). The Weissenberg single-crystal method of X ray diffraction was used to obtain a pattern from a single broken spheroid about 80 micrometers in diameter. A spotty powder pattern was the result, and it confirmed that the spheroid fragment is magnetite and is a crystal aggregate. In preparing the spheroids for diffraction studies, we found them to be very fragile, many breaking on contact with a glass fiber during mounting. This fragility suggests that they would not survive erosion and transport in the sedimentary environment, which indicates a nondetrital origin.

Discussion

The paleomagnetic pole reported in this paper falls near the well established Pennsylvanian-Early Permian segment of the North American APW path. However, the position of this pole alone does not preclude a Late Cambrian age for the magnetization. Several paleomagnetic studies have indicated a Middle to Late Cambrian APW loop [Watts et al., 1980] that includes poles falling near late Paleozoic poles (Figure 5). Although legitimate questions can be raised about the reality of this loop in view of remagnetization possibilities [Gillett, 1982], the negative conglomerate test described earlier presents compelling evidence that the observed magnetization in the Bonneterre is post brecciation and in all likelihood post Cambrian. Moreover, the ages of the rocks that gave paleopoles defining the Cambrian loop also appear to preclude a Late Cambrian age for the age of the magnetization in the Bonneterre. Formations with an age equivalent to that of the Dresbachian Bonneterre [Cook and Bally, 1975] were studied by Watts et al. [1980]; these formations are the Cap Mountain and Lion Mountain of the Llano Uplift, Texas, which gave poles at 33°N, 140°E and 24°N, 146°E. The pole of the overlying Franconian Point Peak Shale of the Wilberns Formation at 6°N, 159°E defines the younging trend of the Cambrian APW loop and it is clear that the Bonneterre pole obtained in this study does not match this late Cambrian segment of the path at all. If any poles on the loop fall near that of the Bonneterre, they all appear to be older than Dresbachian on the basis of stratigraphy (e.g the Rome and Waynesboro formations), radiometric ages of 525 to 530 Ma, or older [Watts et al., 1980]. Comparison of the Bonneterre pole to known post-Cambrian poles for cratonic North America [Van der Voo, 1981] indicates that the Bonneterre magnetization must have occurred during the late Paleozoic, probably during the Pennsylvanian or perhaps the Early Permian.

This Bonneterre secondary magnetization, and also that of the Helderberg carbonates of the eastern United States, has been interpreted as chemical remanent magnetizations (CRM's) residing in magnetite of diagenetic origin [McCabe et al., 1983]. Available paleotemperature estimates [Roedder, 1977] give temperatures too low for this secondary magnetization to be a thermal remagnetization of preexisting magnetite. Blocking temperature and coercivity

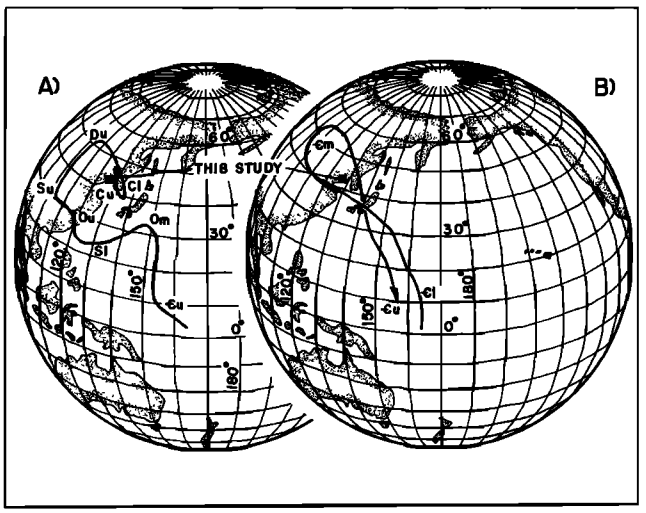


Fig. 5. Pole position derived from the overall mean direction (Table 1) compared with (a) the North American apparent polar wander path [Van der Voo, 1981] and (b) a Cambrian loop proposed by Watts et al. [1980].

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TABLE 2. Conglomerate and Fold Test Results

Site	Test	N	Field Coordinates		Bedding Coordinates		k_1/k_2
			k_1	α_{95}	k_2	α_{95}	
St. Joe #28 Mine	Conglomerate	13	66.1	5.1	13.2	11.9	5.00*
Brushy Creek Mine	Conglomerate	14	30.4	7.3	22.4	8.6	1.36
Buick Mine	Fold	15	21.3	8.5	18.6	9.1	1.14

N is the number of samples studied for the test, k and α_{95} are the statistical parameters [Fisher, 1953] associated with the mean direction, and k_1/k_2 is the ratio of the precision parameters before and after correction for bedding, respectively.
* This ratio confirms a negative conglomerate test for this site, indicating the secondary nature of the magnetization.

spectra of the characteristic magnetization are consistent with magnetite, and this interpretation is strengthened by the rock magnetic results. Magnetic extracts contain abundant magnetite spheroids that we have shown are S, Ti, and Ni free. Since the only magnetites identified in the extracts are either spheroids or spheroid fragments, the stable magnetization can be inferred to reside in these spheroids. The purity of the spheroids is particularly important in inferring their origin. Though the spheroids resemble some volcanic and cosmic magnetic particles, the absence of detectable Ti and Ni in all those spheroids analyzed strongly suggests a low-temperature origin. Pure biogenic magnetite has been found in bacteria [Blakemore, 1975] and some vertebrates [e.g., Kirschvink, 1983] , but the morphology of the spheroids in the Bonneterre makes a bacterial origin unlikely, and the age of the Bonneterre precludes a vertebrate origin. The similarity of the Bonneterre spheroids to diagenetic framboidal pyrite suggests a diagenetic origin, and indeed such an origin best explains all of the observations.

We shall now examine the temporal relationship between the magnetization discussed in this paper and the mineralization of the Viburnum Trend. Though we do not fully understand the processes responsible for the formation of the magnetite spheroids, it has been shown that the magnetite that forms them crystallized in the Pennsylvanian or Early Permian, long after initial sedimentation and after the development of the solution collapse breccias. Also, there is evidence to suggest that magnetite development was a widespread event that may have had no direct connection with MVT mineralization processes that produced ore very locally because the magnetizations associated with the magnetite occur in Bonneterre rocks far removed from known ore deposits as well as close to and within the ore. Similar spheroidal magnetite with the same directions of magnetization have been reported in Paleozoic sedimentary carbonates in other regions devoid of MVT mineralization [McCabe et al., 1983; Scotese et al., 1982] and, though we do not yet have a grasp on the pervasiveness of this event, the available evidence suggests that this was a Pennsylvanian to Early Permian event that affected older sedimentary carbonates over a very wide area of the North American craton.

For economic geologists, a critical question is the age of MVT mineralization relative to the age of the magnetization of the host rock of the Viburnum ores. Since this study has not directly dated the ore or gangue minerals, the relative age of mineralization can only be inferred by logical arguments. Clearly, there are only three possibilities, that the mineralization is the same age, older than, or younger than the remagnetization event, and we shall assess these possibilities in that sequence.

Though it would be pleasing to claim that the mineralization and the magnetite spheroid development were the same age, and thus that we have dated the mineralization directly, several facts deny this. The chemistry of the Viburnum ore fluids, specifically their pH, fO_2, and temperature, was such that they consistently formed sulfides, not oxides, of the metals [Sverjensky, 1981]. Marcasite and/or pyrite are ubiquitous in the ores, whereas hydrothermal magnetite or hematite have never been observed. In fact, many of the ore bodies are bordered by a selvage or halo of marcasite where MVT fluids reacted with their wall rocks [e.g., Sweeney et al., 1977; Rogers and Davis, 1977]. Advocates of a fluid mixing model for MVT deposits might argue that the sulfide ore assemblages were produced by one fluid or fluid mixture of relatively low fO_2 and/or low pH while, simultaneously, another fluid or fluid mixture (perhaps resident meteoric water) was producing magnetite in the wall

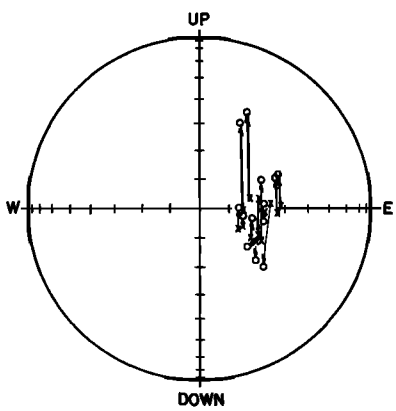


Fig. 6. Conglomerate test from the St. Joe #28 Mine. Crosses represent directions in breccia blocks before bedding correction (k_1 = 66.1, alpha-95 = 5.1°), circles represent these directions after individual bedding corrections have been applied (k_2 = 13.2, alpha-95 = 11.9°). Note the vertical orientation of the projection for clearer display of these nearly horizontal directions.

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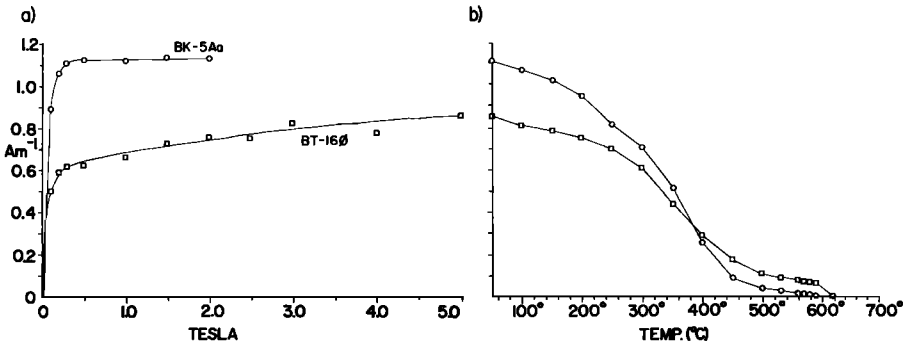


Fig. 7. IRM acquisition and demagnetization results from Bonneterre surface (BT-16O) and mine (BK-5Aa) samples. (a) IRM acquisition. (b) Thermal demagnetization behavior of the IRM of the same samples.

rocks, either by direct precipitation or by replacement of precursor pyrite framboids. Such a construct, though awkward, might be acceptable were it not for the occurrence of the same magnetite in the carbonate breccia blocks within an ore mineral matrix. For these reasons, and also because similar magnetite spheroids are known in far-removed localities having no known association with MVT mineralization [McCabe et al., 1983], we are led to conclude that the Viburnum ores and the magnetite present in their host rocks were not simultaneously formed.

If they are not the same age, could the magnetite spheroids be younger than the MVT mineralization? Here, we have even more serious geochemical difficulties. This postulate would require that the sulfide ore bodies of the Viburnum Trend be formed in place and then survive a pervasive incursion of fluids with a different chemistry which remagnetized rocks throughout the region but left the ore bodies essentially untouched. Were the magnetite spheroids found only in host rocks well removed from ore in the district, one might construct some geochemical scenario that would isolate local sulfide ore bodies from such a regional oxidizing event, but it is the magnetite-bearing breccia blocks within the ore that present an insurmountable problem. Even to reach these blocks, a late fluid would have to pass through the ores, and it seems inconceivable that they could do so without producing at least local and visible alteration of original pyrite and marcasite in the ores to magnetite. Such alteration would be plainly detectable and yet has never been observed in many careful studies of the Viburnum deposits.

If this train of thought is correct, we are left with the single possibility that the magnetite development and its associated paleomagnetic signature preceded MVT mineralization in the Viburnum Trend. Since the magnetite development is known to be Pennsylvanian-Early Permian, this would place the mineralization in Pennsylvanian or post-Pennsylvanian time.

Though some have argued for a much earlier time of mineralization in the Viburnum Trend [e.g., Davis, 1977], the current trend of thought seems to be toward a late Paleozoic age compatible with the paleomagnetic inferences drawn here. Sharp [1978], for example, has proposed a hydrological model of MVT mineralization wherein the MVT deposits of Missouri were formed by deep brines from the Ouachita Basin that were compactionally driven to the shallow scene of ore deposition in Pennsylvanian time. Ohle [1980] has made the salient point that the Ozark Dome as we know it today, a critical element for many in the interpretation of Missouri MVT deposits and their zonation, did not come into existence until late Paleozoic times.

Viewed on a broader stratigraphic scale, but still within the confines of southern Missouri, it is noteworthy that the lead-zinc-barite mineralization of the MVT type occurs in all formations from the Cambrian Lamotte sandstone to Pennsylvanian shales [Snyder, 1967]. Thus, one appears to be dealing either with a single, late mineralization event in Pennsylvanian or later time that affected all existing, older sedimentary rocks or with multiple events of a similar nature that repeated themselves through almost the full span of Paleozoic time. The authors prefer a single Pennsylvanian or post-Pennsylvanian event because of its relative simplicity and because, at least for the one Missouri district studied here, we obtain evidence for a Pennsylvanian or later age for mineralization in the oldest of MVT Paleozoic hostrocks.

Although we have no data to contribute to the age of other North American MVT deposits, one cannot help but to compare the Missouri situation with the North American scene in general. Viewed collectively, the classical MVT deposits of the United States occur in host rocks that span the full Cambrian to Pennsylvanian interval of Paleozoic time and yet, while debate on timing persists locally, the only hard-and-fast fact known in all districts is that the mineralization is younger than lithification of the host rocks in which it occurs. It is therefore interesting to conjecture on the possibility that all of the classical North American MVT deposits were formed more-or-less simultaneously during an event or events of cratonic dimensions. If this proposal is correct, then the age of this major event(s) would have to be at least as young as the youngest rocks hosting MVT ores. That age is Pennsylvanian or post-Pennsylvanian, and we are led, again, to the probable importance of late Paleozoic events in the development of MVT deposits, but this time on a very wide geographic scale. Following this admittedly conjectural thread, one is finally led to ask what was happening in Pennsylvanian or later Paleozoic time that might have contributed to widespread generation of MVT deposits. This was a time when virtually all of the host rocks or their deep basinal equivalents were blanketed by thick Pennsylvanian shale sequences which, though now removed by erosion, may have served as a potential metal source and/or as a compactional driving agent for MVT brine migration. This was also a time of widespread tectonic activity that may well have provided the driving force for updip migration of metalliferous brines that collected in preexisting Paleozoic basins in pre-MVT time. The authors cannot sort out the relative importance of

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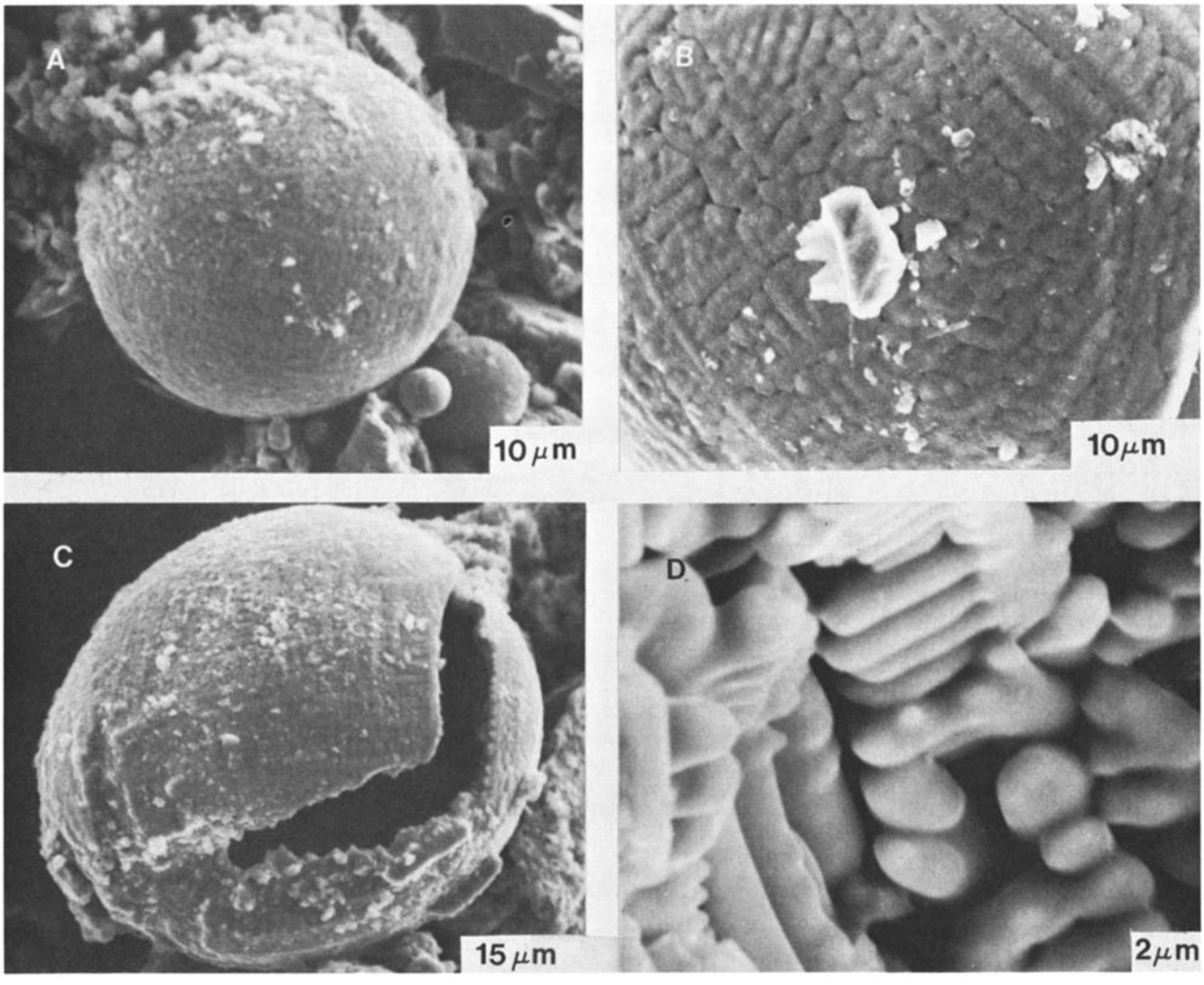


Fig. 8. Scanning electron micrographs of magnetite spheroids extracted from the Bonneterre. (a) Spheroid of pure magnetite from the Viburnum Trend. (b) Close-up of a spheroid showing the surface texture. (c) A cracked, hollow spheroid. (d) Close-up of the interior surface of a broken, hollow spheroid, showing the texture of the inner surface.

these late Paleozoic events but do feel that they should be important considerations for the origin of MVT deposits such as those that we have studied in the Viburnum district.

The issues raised in this paper demand further study. For paleomagnetists the extent and causes of the pervasive remagnetization of Early and Middle Paleozoic carbonates surrounding the North American craton must be more fully investigated. For economic geologists the results presented here suggest a means for dating and better understanding the age of mineralization of classical MVT deposits. Two kinds of follow-up research now seem essential. First, we should go back to the Viburnum Trend with more detailed studies of the paleomagnetism of the gangue carbonates and integrated mineralogical/paleomagnetic studies of the breccia blocks versus the ores that surround them. Second, we feel that paleomagnetic studies of this kind must be extended to other MVT districts because they provide the only means currently available for dating mineralization and its relationship to various diagenetic and tectonic features of the MVT host rocks. As stressed by Ohle [1980], we cannot interpret such mineralization until its age is known and until it can be placed accurately into the full context of basinal evolution.

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